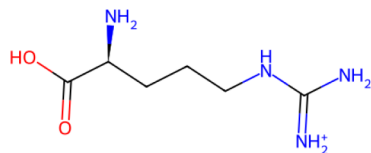
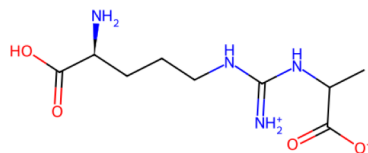


Arginine adducts: identical side-chain charge, differing donor count



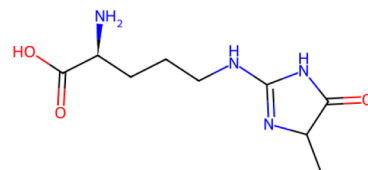
L-Arginine

charge +1 · 5 N-H
native



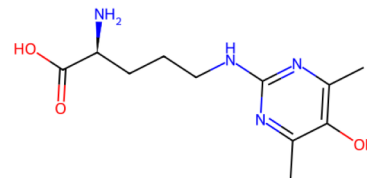
CEA

charge +0 · 4 N-H
 Δ +0.43



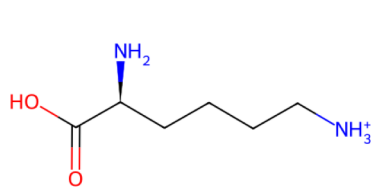
MG-H1

charge +0 · 2 N-H
 Δ -0.48



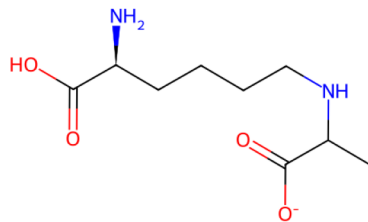
Argpyrimidine

charge +0 · 1 N-H
 Δ -1.13



L-Lysine

charge +1 · 3 N-H



CEL

charge -1 · 1 N-H

Site hydrogen bonding tracks remaining N-H donors, not formal charge: all three arginine adducts are neutral, yet CEA raises it where MG-H1 and argpyrimidine lower it (Table 7).